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MRP: Mini Research Problem Handwritten Digit Recognition Using LeNet

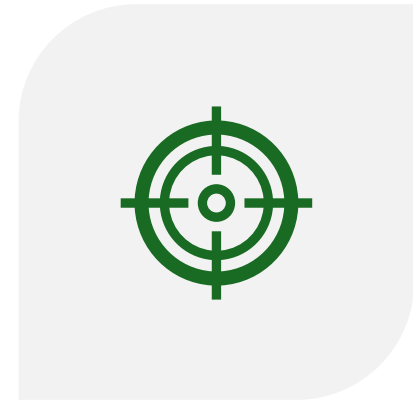
TL;DR - Handwritten Digit Recognition Using LeNet



MOTIVATION: AUTOMATE HANDWRITTEN DIGIT RECOGNITION FOR POSTAL MAIL, BANK CHECKS, AND DOCUMENT DIGITIZATION.



MAIN IDEA: TRAIN THE LENET-5 CONVOLUTIONAL NEURAL NETWORK ON THE MNIST DATASET TO CLASSIFY HANDWRITTEN DIGITS (0-9).



RESULTS: LENET TYPICALLY ACHIEVES ABOUT 99% TEST ACCURACY ON MNIST WHILE REMAINING COMPUTATIONALLY EFFICIENT

Literature Review



McCulloch & Pitts
(1943): First
mathematical neuron
model.



Rosenblatt (1958):
Perceptron introduced
supervised learning.



LeCun et al. (1998):
LeNet-5 demonstrated
CNNs for handwritten
recognition.



Krizhevsky et al. (2012):
AlexNet popularized
deep CNNs.

Approach Used

- Dataset: MNIST (70,000 grayscale images, 28×28).
- Preprocessing: Normalize images to [0,1].
Architecture: Conv → ReLU → AvgPool → Conv → ReLU → AvgPool → Fully Connected → Softmax.
- Loss: Cross-Entropy.
- Optimizer: Adam.
- Evaluation: Test accuracy



Relevant Links

MNIST: <https://yann.lecun.com/exdb/mnist/>

PyTorch MNIST Example:

[https://github.com/pytorch/examples/tree/main/m](https://github.com/pytorch/examples/tree/main/mnist)

nist PyTorch Docs: <https://pytorch.org/> LeNet

Paper:

<http://yann.lecun.com/exdb/publis/pdf/lecun-98.pdf>



Code

- Reference implementation:
<https://github.com/pytorch/examples/tree/main/mnist>

Example personal repository
format:

- <https://github.com//mnist-lenet>